

Rifah Nanzeeba

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EDUCATION

Bangladesh University of Engineering and Technology

Bachelor of Science in Electrical and Electronic Engineering

Major in Communication and Signal Processing

Dhaka, Bangladesh

Feb. 2020 – Mar. 2025

Holy Cross College

Higher Secondary Certificate in Science

Dhaka, Bangladesh

July 2017 – July 2019

Holy Cross Girls' High School

Secondary School Certificate in Science

Dhaka, Bangladesh

January 2015 – May 2017

WORK EXPERIENCE

Trainee Engineer (Intern)

Decor Interior Limited (decor-bd.com)

14 September 2025 – 13 January, 2026

Dhaka, Bangladesh

- Design, Drawing & Simulation of Solar Power Plants and Solution
- New Business Lead Generation & Client Management
- Government Solar Project Document Preparation

RESEARCH PUBLICATIONS

1. ML Based Malware Detection Using Hardware Assisted Performance Counters

IEEE ICCIT'2025 - Published

Authors: Rifah Nanzeeba, Mohammed Ismail Chowdhury, Pran Kanai Saha

- * Developed an HPC-based malware detection framework that assigns classifier weights analytically using recall-focused performance metrics and aggregates predictions through probability-based voting to produce an interpretable, confidence-guided decision.

2. DisBot: An Automated Medicine Dispensing Robot for Efficient Delivery

IEEE QPAIN'2025 - Published

Authors: Sadad Hasan, Kazi Mubashir Tazwar, Nayeemul Islam, Rifah Nanzeeba, Md. Hasan Newaz

- * Designed and implemented a low-cost autonomous medicine dispensing robot integrating RFID and IoT technologies enabling real-time, patient-specific medication delivery with high accuracy and minimal human intervention.

3. IR-Remote Driven and Automated Light Intensity Control System Responsive to Occupancy Level

IEEE QPAIN'2025 - Published

Authors: Tanzim Al Mahmud, Asikur Rahman Jowel, Farhan Zawad Chowdhury, Mehedi Hasan, Rifah Nanzeeba, Imtiaj Ahmed Fahim

- * Designed and implemented a microcontroller-based TRIAC dimming system with IR sensor-based occupancy detection to enable automated, energy-efficient indoor lighting with manual control capability.

4. Pipeline for Real-Time Drowsiness Detection Using Pre-Trained CNN Model

IEEE QPAIN'2025 - Published

Authors: Sadad Hasan, Kazi Mubashir Tazwar, Prantik Paul, Rifah Nanzeeba

- * Designed a real-time driver drowsiness detection pipeline using InceptionV3-based transfer learning for eye-state classification, evaluated on mixed real-world and augmented datasets with 89.98% accuracy and validating performance under low-light and motion-blur conditions.

ML Based Malware Detection Using Hardware Assisted Performance Counters

Supervisor: Dr. Pran Kanai Saha

TECHNICAL SKILLS

Languages: Python, C

Simulation & Power Tools: MATLAB, AutoCAD, ORCAD PSpice, PVSyst, CYME PSAF

Embedded & FPGA: Arduino, ESP32, Keil µVision, Quartus

Domains: Machine Learning, Embedded and IoT Systems, Computer Vision, Automation and Control Systems

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow

Systems & Tools: Linux (Ubuntu), LXD Containers

RESEARCH INTERESTS

Cybersecurity and System Security

Embedded & IoT System and Sensing System in Real Life Applications

Machine Learning and Data Analytics in Real Life Applications

SELECTED PROJECTS

1. Bangla Number-plate of Vehicles Detection with Image Processing Using MATLAB

Developed a template-matching based image processing system using a custom Bangla digit dataset for automatic number plate recognition.

2. IoT Based Aquaculture Monitoring System

Designed an IoT-based monitoring and control system for temperature, pH, water level, and automated feeding in aquaculture environments.

3. Smart Dual Axis Solar Tracker with Automated Cleaning Mechanism

Developed a dual-axis solar tracking system using LDR sensors for optimal solar panel alignment, with an automated cleaning mechanism to maintain efficiency.

4. Induction Motor Characteristics Performance Analysis Using MATLAB

Developed a MATLAB App Designer tool to estimate motor parameters and simulate torque-speed, power, loss, and efficiency characteristics.

REFERENCE

Dr. Pran Kanai Saha

Professor

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Dr. Apratim Roy

Professor

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